

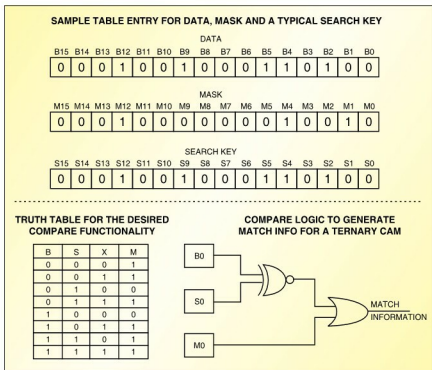


Fig. 2: Working of a ternary CAM cell

host was residing in a neighbouring network, the router will have to look-up into its routing table to identify the neighbouring router address that needs to be used to reach the desired destination.

Hosts are statically located in two distant networks. The router at the edge of the network on the source host side will try to find the destination address or the corresponding router next-hop address. If it does not find it, the router decides to use the default gateway which can be the router at the service provider's edge, whose job would be to resolve the best route to reach the destination.

While one host is statically located in a network, the other host is mobile. In this case, it would be up to the routers of the network in which the host is switching to, to update their tables and also inform the neighbouring routers about addition or deletion of the hosts.

In the above scenarios, we see that the look-up table needs to be updated regularly to keep tab on the latest happenings on the network, and based

on the information residing on these look-up tables, the packets are communicated between hosts.

Working of a CAM

Let us see how a CAM works in hardware. As mentioned earlier, a CAM can be either binary or ternary.

Binary CAM cell. CAM logic, as shown in Fig. 1, will simply have the storage circuitry (an SRAM or a DRAM cell) and comparator logic to provide the match information for the searches performed. In this case, a XOR gate is used to compare between the stored data and the search data.

Ternary CAM cell. A ternary CAM, as the name indicates, has three states: 0, 1 and 'X'. A "don't-care" condition exists in a ternary CAM to mask-off certain bits for a partial match.

Structurally, a ternary CAM looks as shown in Fig. 2. In the context, two sets of data word appear for a single entry in the table: one referring to the data to be looked-up for and the other referring to the mask that is to be applied on every bit during look-up. The convention used w.r.t. the mask in this

context is: "A '1' refers to data masked and a '0' refers to data not being masked." So in the example depicted here, bits B1, B4 and B12 of data are masked-off using the mask-bits from the mask word. Effectively, a match would occur.

The possibility of masking-off of certain bits of the data opens up a wide range of applications where partial match is necessary; for instance, at the edge of a router, where the router only worries about the destination network address and not the host ID. (An IP address consists of two fields: network ID and host ID.)

Typical CAM usage

Typically, a network board consists of a network processor, associated memories and CAMs (multiple CAMs used in cascade configuration when the table density is more), and the physical interface.

As shown below, the network processor has multiple interfaces:

1. **Ingress/Egress ports.** Ports through which the data exchange happens

2. **Host interface.** This interface is used to access the network processor for configuration setting and other updates to be carried out

3. **Associated SRAM interface.** The associated SRAM holds the packet forwarding information such as the port numbers and QoS, which is accessed by the network processor to process the packet received. The index or address for accessing the associated SRAM is typically driven by the search result received from the CAM

4. **CAM interface.** This is used to access the CAM, update the database, perform searches and access the results. The result of the searches performed (index for a search) is typically used by the network processor to link it to the associated SRAM address in order to access the forwarding related information

As seen here, the index returned by the CAM is directly linked to the associated SRAM. So some CAMs come